



General Physics II

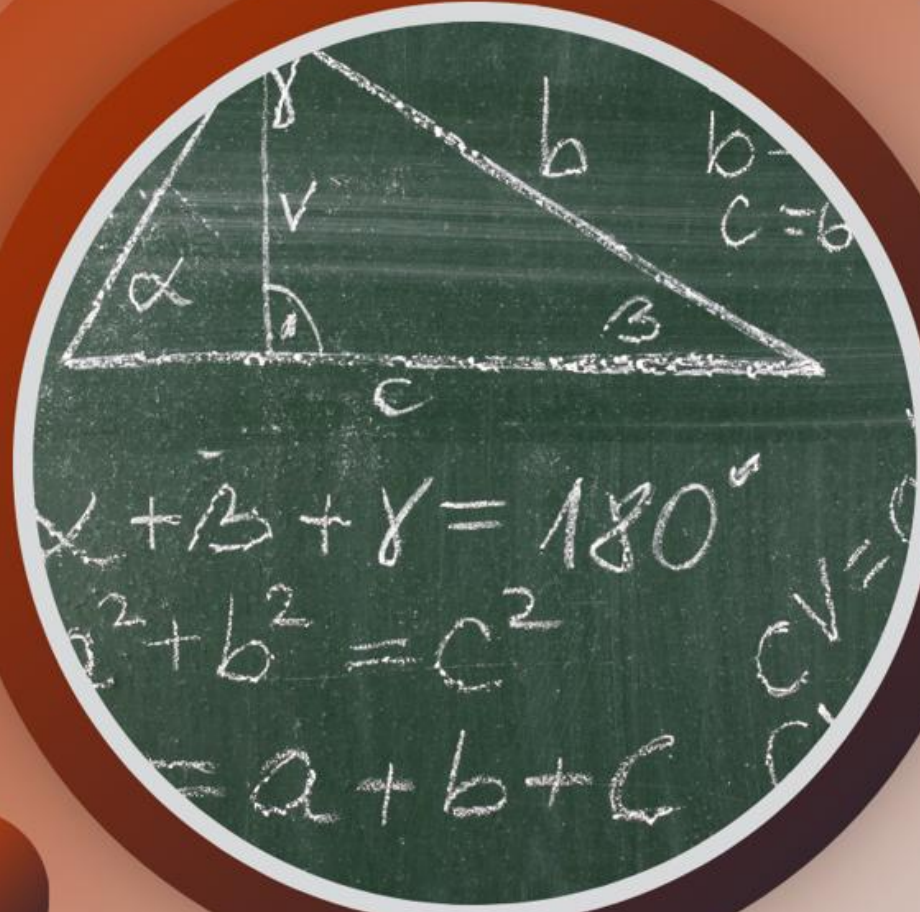
PHY121

College of Science and Technology



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MODULE 6: Maxwell's Equations & Electromagnetic Waves

UNIT 1: Maxwell's Equations

Learning Objectives

- 1 Identify the gap in Ampere's law that Maxwell discovered
- 2 State all four Maxwell's equations and their physical meaning
- 3 Derive the wave equation and show EM waves travel at c
- 4 Explain why light is an electromagnetic wave

"In the beginning, God said, the four-dimensional divergence of an antisymmetric second-rank electromagnetic tensor equals zero, and there was light."



Quick Poll

Which of the four laws do you think has a problem?

A

Gauss's law for electric fields

B

Gauss's law for magnetism

C

Faraday's law

D

Ampere's law



Quick Poll

Which of the four laws do you think has a problem?

A Gauss's law for electric fields

B Gauss's law for magnetism

C Faraday's law

D Ampere's law

Ampere's Law (As You Learned It)

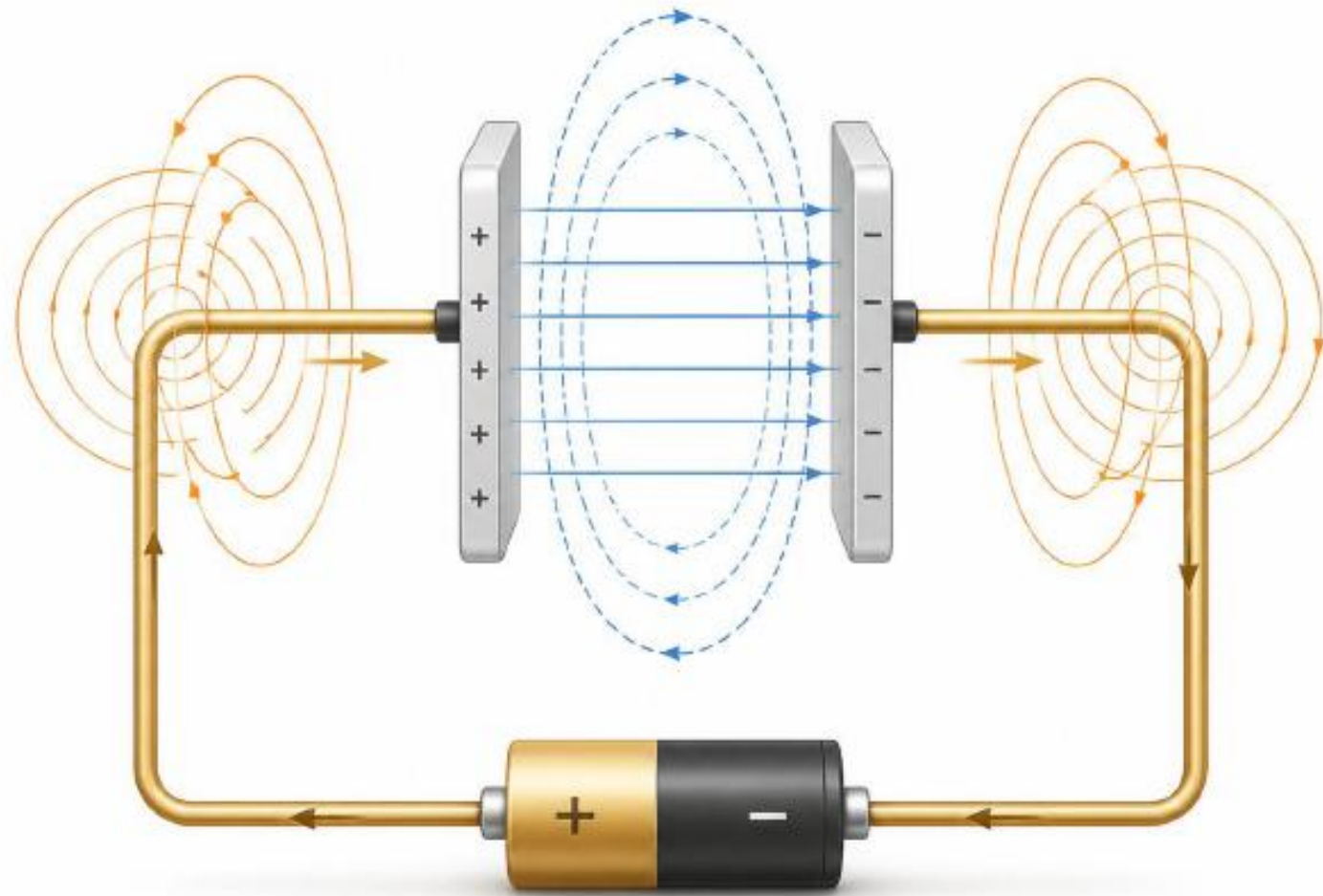
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

This works perfectly for:

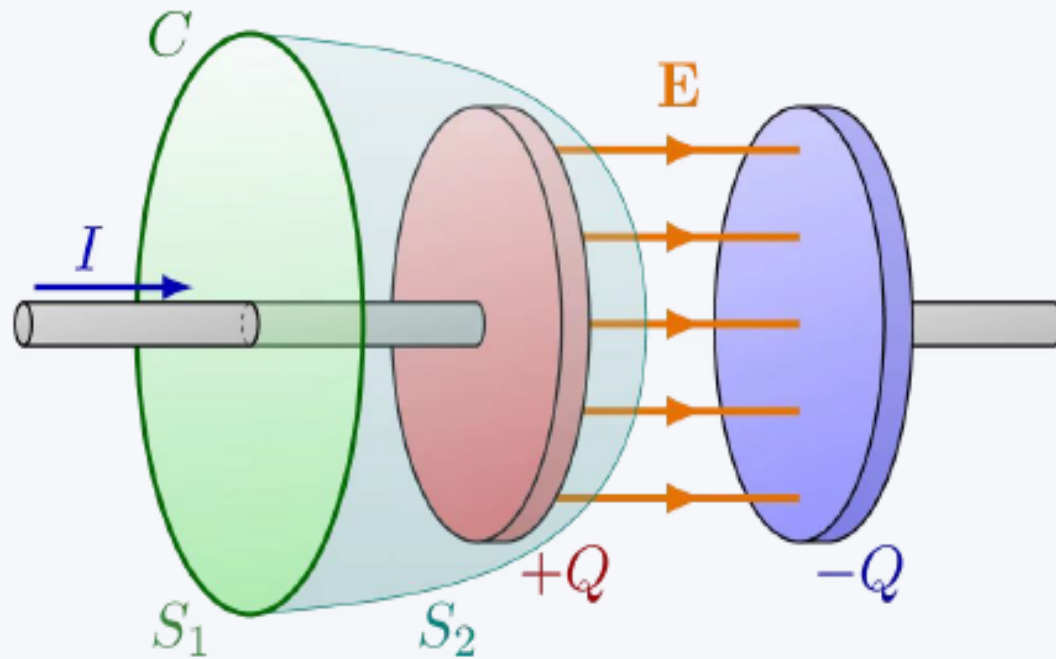
- ✓ Long straight wires
- ✓ Solenoids
- ✓ Toroids
- ✓ Any steady current

 **Think-Pair-Share**

This law worked for every problem you solved. So what could possibly be wrong with it?



The Charging Capacitor Problem



Surface 1: Through the wire

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} \checkmark$$

Surface 2: Between the plates

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} \times$$

Same loop. Same law. Two different answers.



LOGICAL CONTRADICTION

Surface 1

(through the wire)

$$\oint \vec{B} \cdot d\vec{l} = \mu^0 I$$

$\neq 0$

VS

Surface 2

(between the plates)

$$\oint \vec{B} \cdot d\vec{l} = 0$$

$= 0$

*A law of physics cannot give different answers
for the same physical situation.*



Quick Poll

What do you think is happening?

A

Ampere's law is completely wrong

B

Something else creates a 'current' between the plates

C

The magnetic field is actually zero there

D

We're applying the law incorrectly



Quick Poll

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Maxwell's Fix: The Displacement Current

The Displacement Current

Maxwell's insight: A changing electric field produces a magnetic field, just like a current does.

Displacement Current:

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Corrected Ampere-Maxwell Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$



Quick Poll

A parallel-plate capacitor is being charged by a constant current of 2 A. The displacement current between the plates is:

A

0 A

B

Less than 2 A

C

Exactly 2 A

D

Greater than 2 A



Quick Poll

A parallel-plate capacitor is being charged by a constant current of 2 A. The displacement current between the plates is:

C

Exactly 2 A

By charge conservation, the displacement current between the plates must equal the conduction current in the wire.



Worked Example

A capacitor with circular plates of radius 10 cm is being charged by a steady current of 2.5 A.

- (a) What is the displacement current between the plates?
- (b) At what rate is the electric field between the plates changing?



Solution

(a) By charge conservation, the displacement current equals the conduction current:

$$I_d = I_{\text{conduction}} = 2.5 \text{ A}$$



Solution

(b) The plate area is $A = \pi r^2 = \pi(0.10)^2 = 3.14 \times 10^{-2} \text{ m}^2$.

We start from the displacement current definition $I_d = \epsilon_0, d\Phi_E/dt$. Between parallel plates, the electric field E is uniform across the plate area A , so the electric flux simplifies to $\Phi_E = EA$. Substituting:

$$I_d = \epsilon_0 \frac{d(EA)}{dt} = \epsilon_0 A \frac{dE}{dt}$$



Solution

Rearranging for dE/dt :

$$\frac{dE}{dt} = \frac{I_d}{\epsilon_0 A} = \frac{2.5}{(8.85 \times 10^{-12})(3.14 \times 10^{-2})} = 9.0 \times 10^{12} \text{ V/(ms)}$$

Interpretation:

The electric field changes at ~9 trillion V/m per second. This enormous rate is what generates a displacement current matching the 2.5 A in the wire.

Maxwell's Equations

- The Complete Picture
- Four equations that contain everything
- about electricity, magnetism, and light.

Maxwell's Equations

Gauss (E)

$$\oint \vec{E} \cdot d\vec{A} = Q / \epsilon_0$$

"Charges create electric fields"

Gauss (B)

$$\oint \vec{B} \cdot d\vec{A} = 0$$

"No magnetic monopoles"

Faraday

$$\oint \vec{E} \cdot d\vec{l} = -d\Phi^I/dt$$

"Changing B creates E"

Ampere-Maxwell

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

"Currents & changing E create B"

Differential Form of Maxwell's Equations

#	Integral Form	Differential Form
I	$\oint \vec{E} \cdot d\vec{A} = Q_{\text{enc}}/\epsilon_0$	$\nabla \cdot \vec{E} = \rho/\epsilon_0$
II	$\oint \vec{B} \cdot d\vec{A} = 0$	$\nabla \cdot \vec{B} = 0$
III	$\oint \vec{E} \cdot d\vec{l} = -d\Phi_B/dt$	$\nabla \times \vec{E} = -\partial \vec{B} / \partial t$
IV	$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0, d\Phi_E/dt$	$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0, \partial \vec{E} / \partial t$

The Beautiful Symmetry

Faraday's Law

Changing $\vec{B} \rightarrow$ creates $\vec{E}$

$$\oint_c \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$



Ampere-Maxwell Law

Changing $\vec{E} \rightarrow$ creates $\vec{B}$

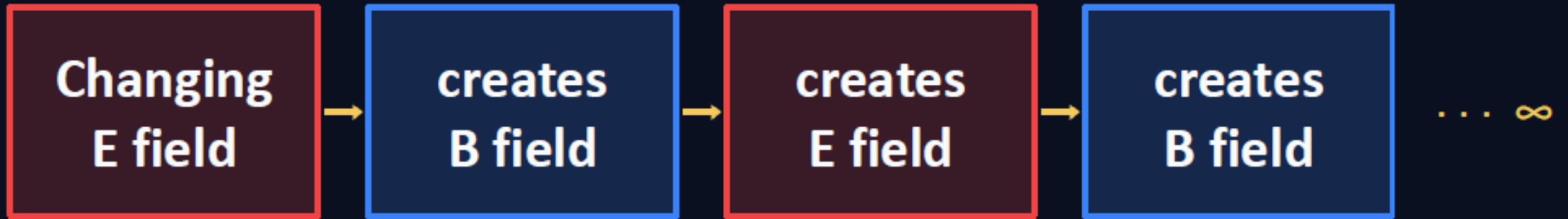
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

This mutual creation is the engine that drives electromagnetic waves.

MODULE 6: Maxwell's Equations & Electromagnetic Waves

UNIT 2: Electromagnetic Waves

The Chain Reaction

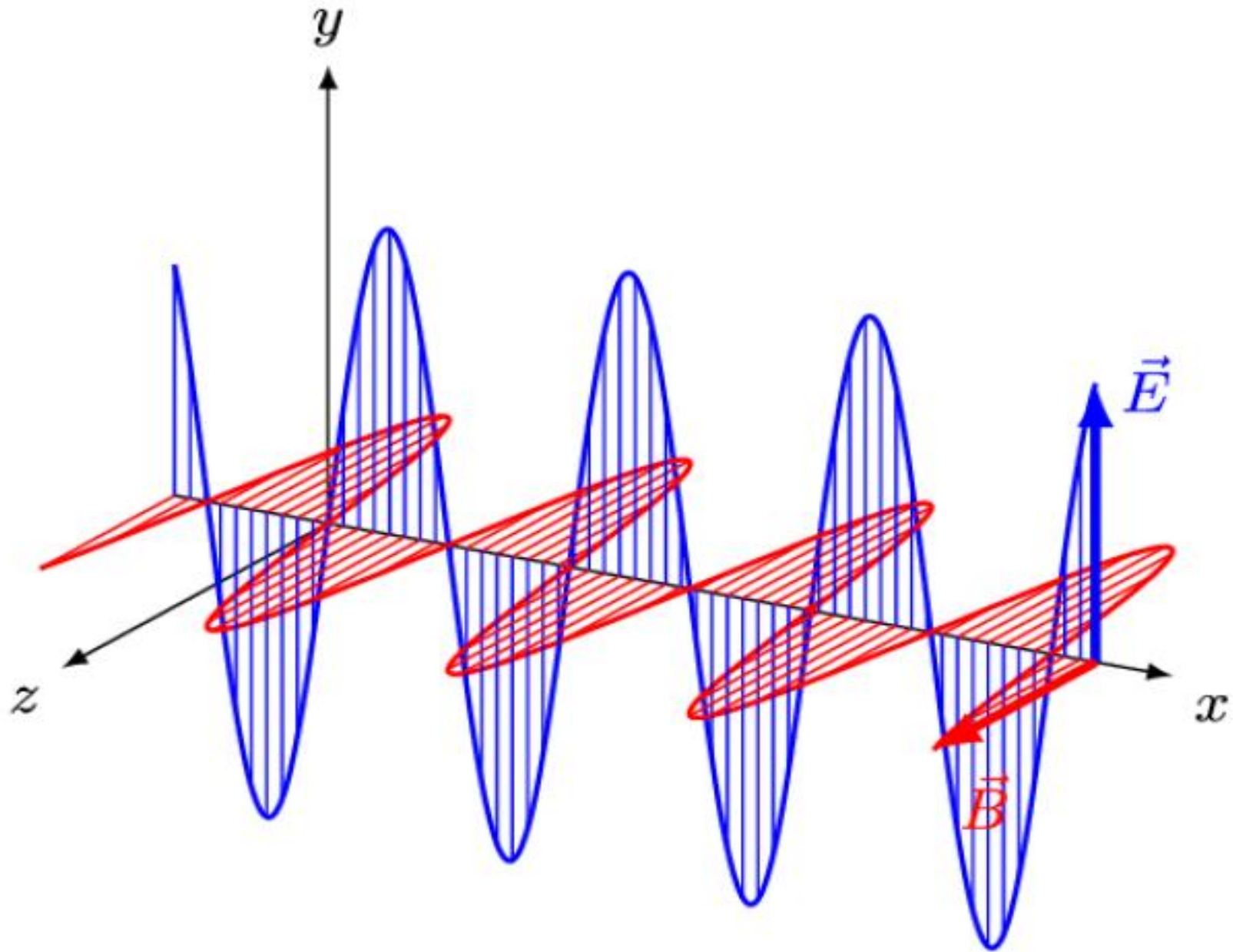


They feed each other.

They sustain each other.

They propagate through space.

That's a **WAVE**.



The Speed of Light

The moment that shook 19th-century physics.

Deriving the Wave Equation

In free space (no charges, no currents), Maxwell's equations give:

$$\frac{\partial^2 \vec{E}}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Comparing with the standard wave equation:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$



$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Let's calculate this number...

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$
$$v = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}}$$

$$\mathbf{v = 2.998 \times 10^8 \text{ m/s}}$$

Speed of light:

$$\mathbf{c = 2.998 \times 10^8 \text{ m/s}}$$

They are exactly the same.

The velocity of transverse undulations in our hypothetical medium, calculated from the electromagnetic experiments... agrees so exactly with the velocity of light... that we can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena.

— James Clerk Maxwell, 1865

He predicted electromagnetic waves from pure mathematics. He never saw one. He never built an antenna. Hertz confirmed them experimentally in 1887, eight years after Maxwell's death.

Properties of Electromagnetic Waves



Transverse

E and B oscillate perpendicular to propagation

c

Speed: $c = 3 \times 10^8$ m/s

In vacuum, all EM waves travel at the speed of light

~

No medium needed

Unlike sound, EM waves travel through empty space

∞

$E = cB$

Electric and magnetic fields are linked at every point

W

Carry energy

Energy flux: Poynting vector $\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$

The Plane Wave Solution

$$\frac{\partial^2 \vec{E}}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

The wave equation is satisfied by any function of the form $f(x - ct)$, but the most useful solution is a **sinusoidal plane wave** travelling in the +x-direction:

$$E_y = E_0 \sin(kx - \omega t)$$
$$B_z = B_0 \sin(kx - \omega t)$$

where:

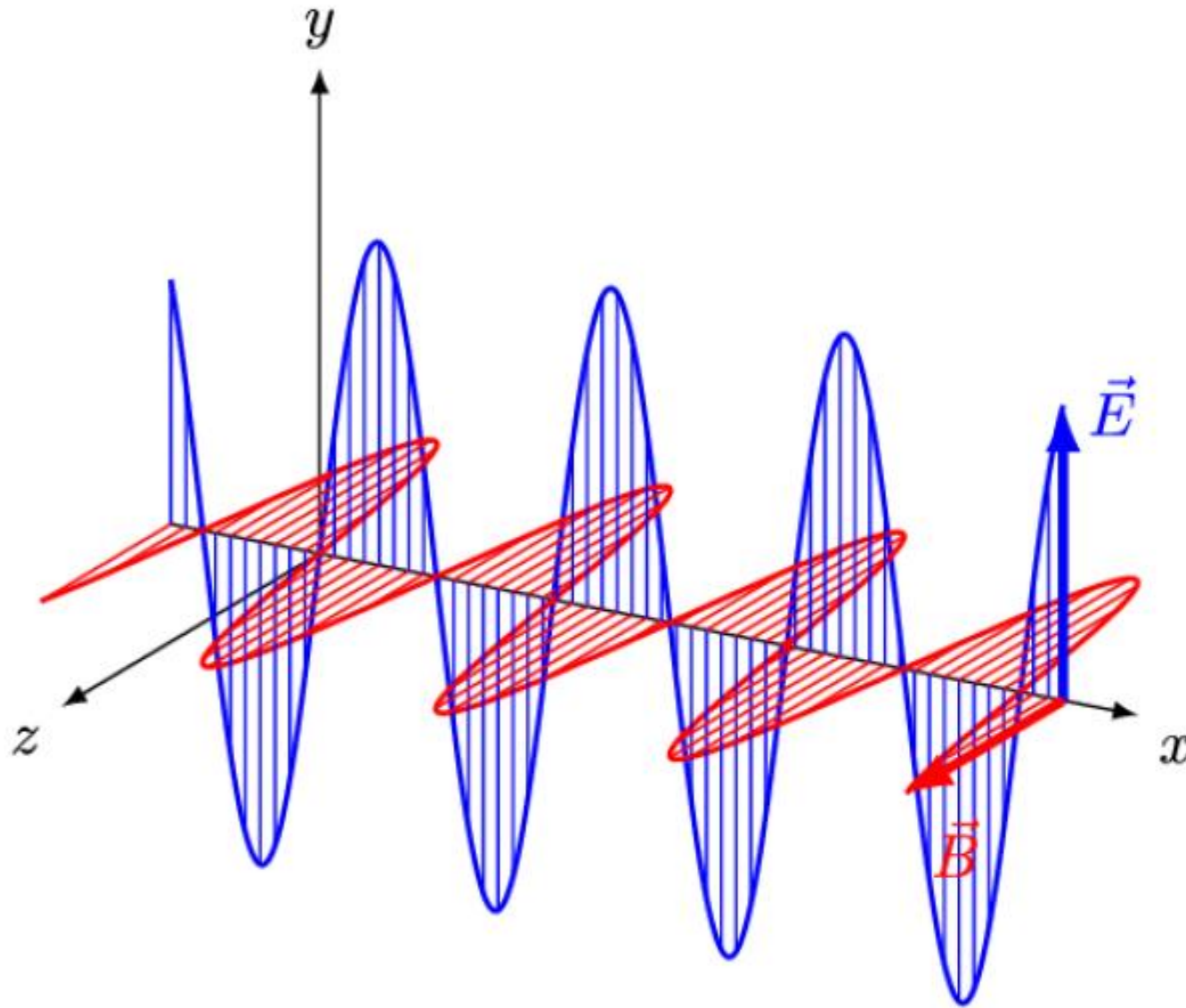
- E_0, B_0 = field amplitudes
- $k = 2\pi/\lambda$ = wave number (rad/m)
- $\omega = 2\pi f$ = angular frequency (rad/s)
- λ = wavelength, f = frequency

The Plane Wave Solution

$$E = cB$$

For every electromagnetic wave, speed, frequency, and wavelength are locked together by:

$$c = f\lambda$$





Worked Example

A radio transmitter produces an electromagnetic wave with an electric field amplitude of 0.080 V/m . What is the corresponding magnetic field amplitude?

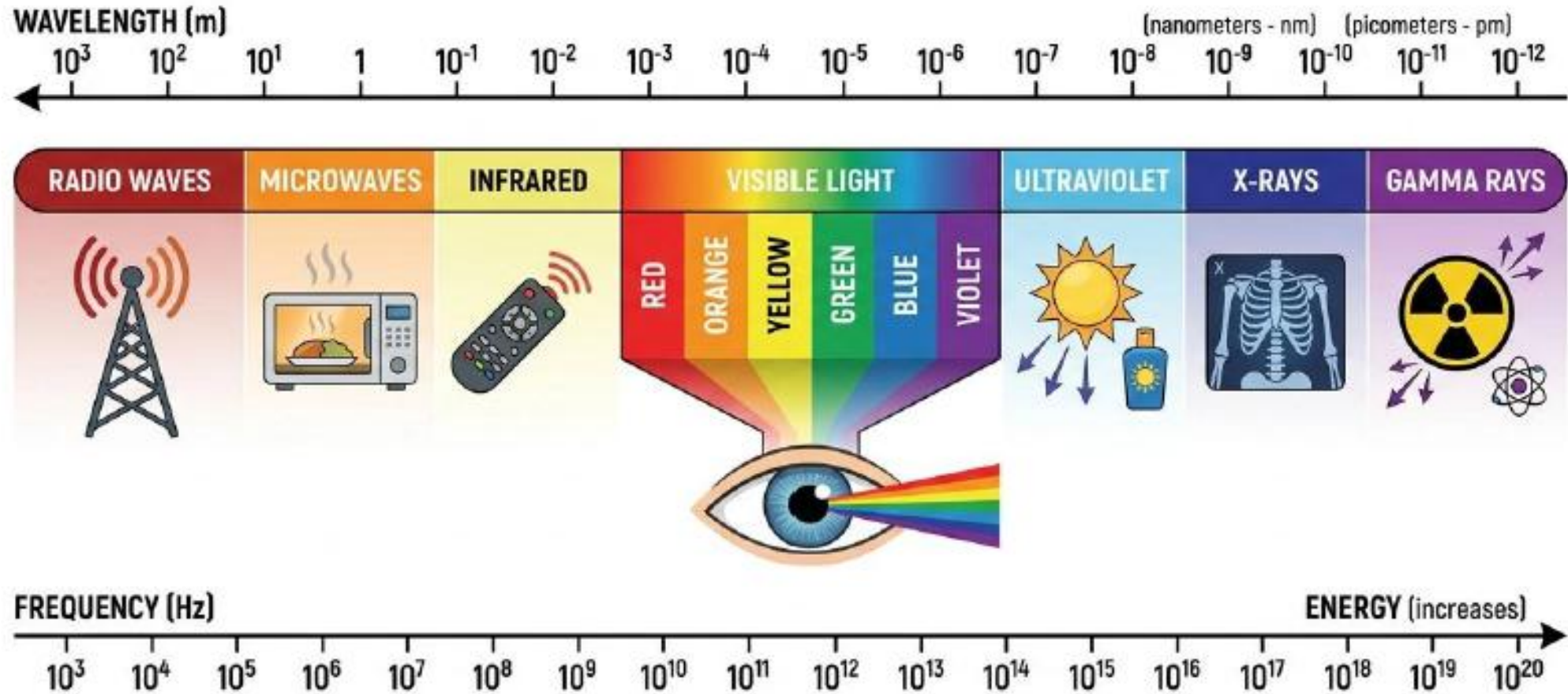


Solution

Using the relationship $E_0 = cB_0$:

$$B_0 = \frac{E_0}{c} = \frac{0.080}{3.0 \times 10^8} = 2.67 \times 10^{-10} \text{ T}$$

THE FULL ELECTROMAGNETIC SPECTRUM



Energy in Electromagnetic Waves

Energy Density

Electromagnetic waves carry energy. The energy per unit volume (energy density) stored in the fields is:

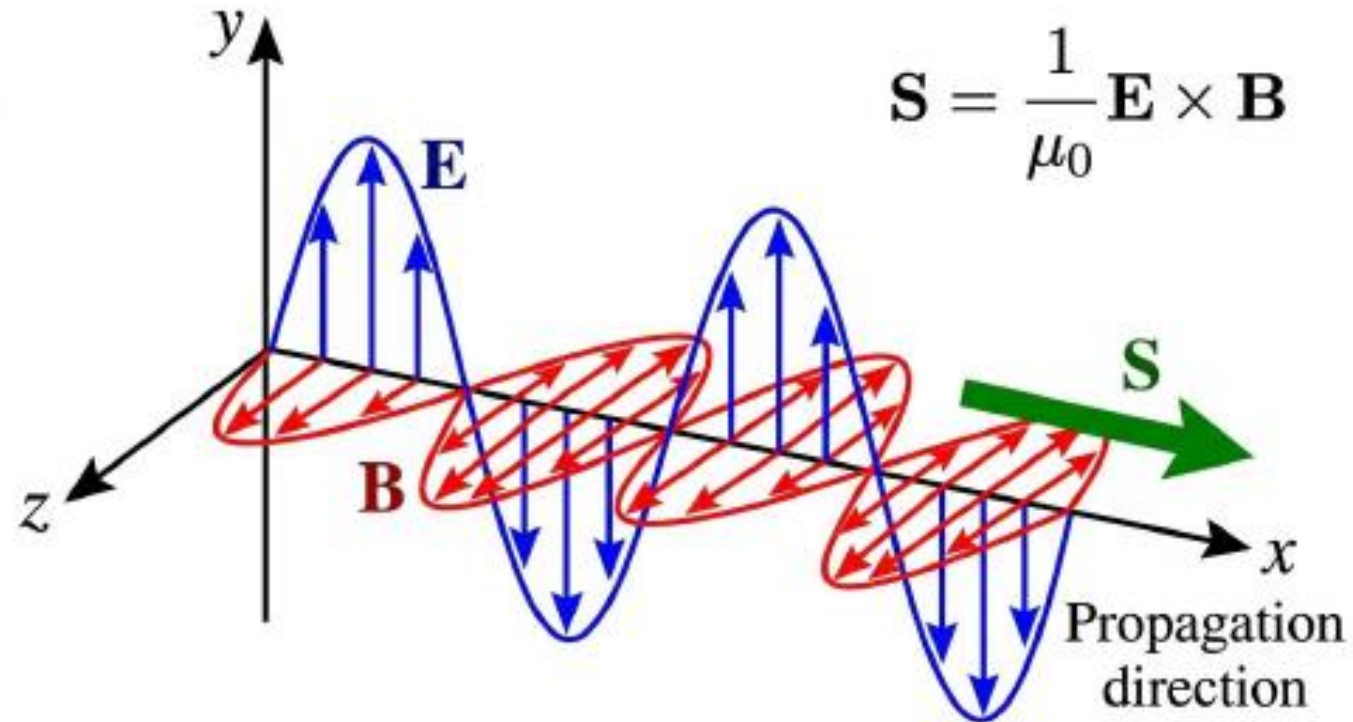
$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2$$

Since $E = cB$ and $c = 1/\sqrt{\mu_0\epsilon_0}$, the electric and magnetic contributions are equal:

$$u = \epsilon_0 E^2 = \frac{B^2}{\mu_0}$$

Intensity (Poynting Vector)

The rate at which energy is transported per unit area is given by the **Poynting vector**:





The **intensity** (average power per unit area) of an EM wave is:

$$I = \frac{1}{2} \epsilon_0 c E_0^2 = \frac{E_0^2}{2\mu_0 c} = \frac{c B_0^2}{2\mu_0}$$

where:

- I = intensity (W/m^2)
- E_0 = electric field amplitude (V/m)
- B_0 = magnetic field amplitude (T)



Worked Example

A laser pointer emits 5.0 mW of power in a beam with cross-sectional area $3.0 \times 10^{-6} \text{ m}^2$. Calculate (a) the intensity and (b) the electric field amplitude of the laser beam.



Solution

(a) Intensity:

$$I = \frac{P}{A} = \frac{5.0 \times 10^{-3}}{3.0 \times 10^{-6}} = 1667 \text{ W/m}^2$$



Solution

(b) From $I = \frac{1}{2} \epsilon_0 c E_0^2$:

$$E_0 = \sqrt{\frac{2I}{\epsilon_0 c}} = \sqrt{\frac{2(1667)}{(8.85 \times 10^{-12})(3.00 \times 10^8)}} = \sqrt{\frac{3334}{2.655 \times 10^{-3}}} \approx 1120 \text{ V/m}$$

Interpretation: Despite its tiny power (only 5 mW), the laser has an intensity higher than sunlight (1000 W/m^2) because its energy is concentrated in a very small area. This is why lasers can be hazardous to the eyes.

Maxwell's Equations in Your Daily Life

Medical Imaging

X-rays (high-frequency EM waves) are used in hospitals for diagnostic imaging.

~1800 MHz

Power Transmission and EM Radiation

High-voltage power lines carrying AC current emit low-frequency electromagnetic radiation.

88-108 MHz

Wi-Fi

The Wi-Fi in this building uses microwave-frequency EM waves

2.4 / 5 GHz

WhatsApp Calls

Voice data encoded as EM waves, routed through cell networks

Mixed

All predicted by a theorist who noticed one equation was inconsistent.

Key Takeaways

1

Maxwell's equations are four equations that completely describe classical electromagnetism. The key addition was the displacement current in the Ampere-Maxwell law.

2

Displacement current ($I_d = \epsilon_0, d\Phi_E/dt$) is the effect of a changing electric field, not a flow of charges. It ensures continuity and symmetry in the equations.

3

Electromagnetic waves arise from the mutual induction of electric and magnetic fields. They travel at $c = 1/\sqrt{\mu_0\epsilon_0} = 3 \times 10^8 \text{ m/s}$ in vacuum.

4

EM waves are transverse: $\vec{E}$, $\vec{B}$, and the propagation direction are mutually perpendicular.
 $E = cB$ at all times.

5

The electromagnetic spectrum encompasses radio waves to gamma rays, all travelling at the same speed but differing in frequency.